Power BI Dashboard Performance Analysis Using Performance Analyzer

1. Introduction

Power BI Performance Analyzer is a built-in tool that helps monitor and evaluate the performance of report visuals. It provides detailed information about the time taken to execute queries, render visuals, and complete other processing activities.

In this exercise, Performance Analyzer was used to examine the Sales Overview Dashboard and understand how efficiently its visualizations perform.

2. Objective

The main objectives of this exercise are:

- Learn how to use Performance Analyzer in Power BI Desktop.
- Measure the execution time of individual dashboard visuals.
- Understand DAX query execution and visual rendering time.
- Identify visuals that take comparatively longer to load.
- Explore techniques to improve dashboard performance.

3. Tools and Technologies Used

- Microsoft Power BI Desktop
- Sales Overview Dashboard
- Performance Analyzer
- DAX (Data Analysis Expressions)

4. Implementation Steps

4.1 Opening the Dashboard

Opened the existing Sales Overview Dashboard in Power BI Desktop to examine the performance of its visualizations.

4.2 Accessing Performance Analyzer

Navigated to the **Optimize** tab in the Power BI ribbon and selected Performance Analyzer to open the performance monitoring pane.

4.3 Starting Performance Recording

Clicked the **Start Recording** option to begin monitoring the execution time of dashboard visuals.

4.4 Refreshing Dashboard Visuals

Selected **Refresh Visuals** to reload the report visuals and capture their individual execution performance.

4.5 Reviewing Performance Details

Examined the execution details displayed for each visual in the Performance Analyzer pane.

The recorded information included:

- DAX Query – Time taken to execute the query.
- Visual Display – Time required to render the visual on the report page.
- Other – Time spent on additional internal processing activities.

4.6 Testing Interactive Filters

Changed the City and Date slicer selections to observe how dashboard performance varied under different filtering conditions.

5. Performance Analysis

Performance Analyzer was used to compare the execution time of individual dashboard components.

The analysis focused on three major performance measurements:

1. **DAX Query Time:** Measures how long Power BI takes to execute the queries required by a visual.
2. **Visual Display Time:** Measures the time required to display and render the visual.
3. **Other Processing Time:** Represents additional processing activities involved in loading the visual.

These measurements help identify which dashboard components consume more processing time.

6. Performance Observations

The execution details of different dashboard visuals were reviewed and compared.

The analysis helped understand that individual visuals may have different loading times depending on their calculations, data volume, and rendering requirements.

Visuals with relatively higher execution times can be investigated further to identify possible performance bottlenecks.

7. Performance Optimization Recommendations

The following practices can help improve dashboard performance:

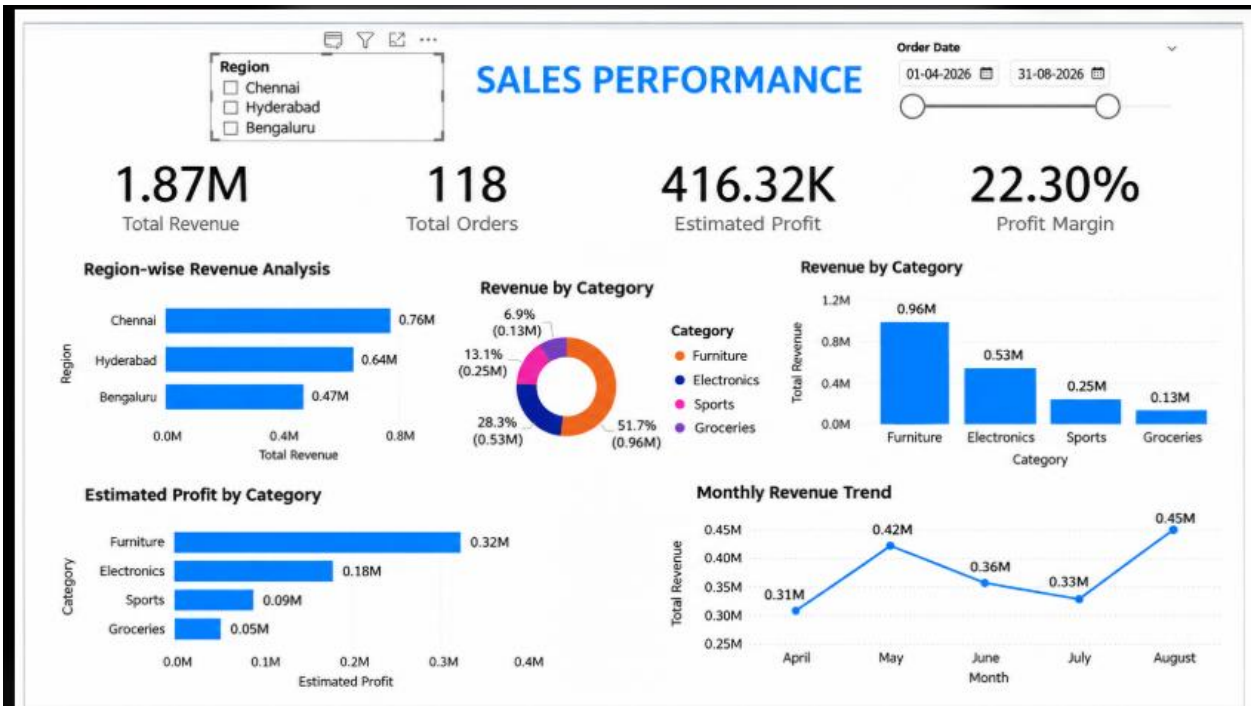
1. **Optimize DAX Measures:** Simplify complex calculations and avoid unnecessary expressions.
2. **Remove Unused Columns:** Keep only the required fields in the data model.
3. **Reduce Visual Overload:** Avoid placing too many visuals on a single report page.
4. **Improve Data Modelling:** Maintain appropriate relationships and data types.
5. **Limit Unnecessary Data:** Reduce excessive data points in charts and tables.
6. **Optimize Filtering:** Use efficient filtering and slicer configurations.
7. **Use Efficient Queries:** Improve query execution by simplifying data transformations and calculations.

8. Conclusion

Successfully explored and used Performance Analyzer in Microsoft Power BI Desktop to evaluate the Sales Overview Dashboard.

The exercise provided practical experience in recording visual execution times, examining DAX query performance, understanding visual rendering time, and comparing performance under different filtering conditions.

It also demonstrated how Performance Analyzer helps identify potential performance issues and provides useful insights for building faster and more efficient Power BI dashboards.


Region
Chennai
Hyderabad
Bengaluru
SALES PERFORMANCE
Order Date
01-04-2026
31-08-2026
1.87M
Total Revenue
118
Total Orders
416.32K
Estimated Profit
22.30%
Profit Margin
Region-wise Revenue Analysis
Chennai
Hyderabad
Bengaluru
0.76M
0.64M
0.47M
Region
0.0M
0.4M
0.8M
Total Revenue
Revenue by Category
6.9%
(0.13M)
13.1%
(0.25M)
28.3%
(0.53M)
51.7%
(0.96M)
Category
Furniture
Electronics
Sports
Groceries
Revenue by Category
1.2M
0.8M
0.4M
0.0M
Total Revenue
0.96M
0.53M
0.25M
0.13M
Furniture
Electronics
Sports
Groceries
Category
Estimated Profit by Category
Furniture
Electronics
Sports
Groceries
0.32M
0.18M
0.09M
0.05M
Category
0.0M
0.1M
0.2M
0.3M
0.4M
Estimated Profit
Monthly Revenue Trend
0.45M
0.40M
0.35M
0.30M
0.25M
Total Revenue
0.31M
0.42M
0.36M
0.33M
0.45M
April
May
June
July
August
Month